

Miray Hirabayashi

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EDUCATION

Iowa State University

Ames, Iowa

Bachelor of Science, Major in Software Engineering, Minor in Cybersecurity

Expected Graduation: May 2027

- **Relevant Coursework:** Object-Oriented Programming, Data Structures, Algorithms, Digital Logic, Cybersecurity Concepts, Embedded Systems, UI Design, Software Development Practices, Database Management Systems, Software Testing

WORK EXPERIENCE

Iowa Department of Transportation

Ames, Iowa

Administrative Intern

May 2025 - Present

- Developed a **Python** CLI application to automate validation and reconciliation between Workday and Microsoft Access datasets, processing **11,000+** records and eliminating manual verification.
- Automated data entry and reporting workflows using **Microsoft Excel (XLOOKUP)**, improving data accuracy and reducing manual processing time.

Iowa State University Language Studies Resource Center

Ames, Iowa

Student Assistant

Dec 2024 - Present

- Provided technical support and troubleshooting for hardware and software used by students and faculty.

TECHNICAL SKILLS

- **Programming Languages:** Java, Python, JavaScript, TypeScript, C, PHP, SQL, Swift, HTML, CSS
- **Frameworks & Libraries:** React, Angular, Next.js, Node.js, Express, Flask, Prisma, Tailwind CSS, SwiftUI, scikit-learn
- **Development Tools:** Git (GitHub, GitLab), Supabase, Xcode, Android Studio, Figma, Postman, Vercel
- **Databases:** PostgreSQL, MySQL, MongoDB
- **Operating Systems & Environments:** Linux (Ubuntu, Debian, Kali), macOS, Windows

PROJECTS

GlobeBase - Interactive 3D Travel Tracker | Next.js, React, TypeScript, Prisma, PostgreSQL, Three.js, Tailwind CSS

- Built a full-stack 3D globe travel tracker rendering **250+** geographic entities using **WebGL (Three.js)** with a hybrid SSR-safe **Next.js** App Router architecture.
- Designed optimistic UI updates with SWR cache synchronization across 4 typed REST API routes validated via **Zod**, enabling instant cross-component state consistency.
- Engineered **Prisma**-based **PostgreSQL** upsert logic to eliminate manual seeding for **195** countries, and integrated **Supabase Storage** to persist user travel state across sessions and devices.

Iowa After Class | Next.js, React, Node.js, TypeScript, Tailwind CSS

- Built and deployed a scalable full-stack event discovery platform with **Next.js** and **TypeScript**; developed backend API routes using **Node.js** to integrate real-time data from the **Ticketmaster Discovery API**.
- Designed a responsive frontend with **React** and **Tailwind CSS**, leveraging AI-assisted prototyping for UI/UX exploration and iterative design improvements.
- Launched on **Vercel** with integrated analytics tracking, reaching **130+** users within the first week and validating early user engagement.

WorkSync - Android App | Java, MySQL, Android Studio, Gradle

- Developed an employee and project management application for small businesses to track staff, assign tasks, and manage workflows.
- Built Android UI and application logic in **Java**, integrating REST APIs, WebSockets, and full CRUD functionality.
- Collaborated in an Agile team using **Git/GitLab** with feature branches and merge requests, implemented CI/CD pipelines, and developed unit tests to improve code quality and deployment reliability.